[INCH-POUND]

A-A-55753
25 September 1995
SUPERSEDING
ZZ-R-765/9
11 April 1991

COMMERCIAL ITEM DESCRIPTION

# RUBBER, SILICONE, TUBING, NONMETALLIC, ROUND, FLEXIBLE, SHAPE 9

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for ZZ-R-765/9, which is cancelled

1. SCOPE. This description covers a silicone rubber nonmetallic round flexible tubing of various classes and grades.

2. CLASSIFICATION. The silicone rubber shall be of the following classes and grades as specified. The designated grade number corresponds to the nominal Shore Durometer A hardness value.

Class 1A - Low temperature resistant
Grade - 40, 50, 60, 70 and 80

Class 1B - Low temperature resistant and low compression set at high temperature
Grade - 40, 50, 60, 70 and 80

Class 2A - High temperature resistant
Grade - 25, 40, 50, 60, 70 and 80

Class 2B - High temperature resistant and low compression set
Grade - 25, 40, 50, 60, 70 and 80

Class 3A - Low temperature, tear and flex resistant
Grade - 30, 50 and 60

Class 3B - Tear and flex resistant
Grade - 30, 50, 60, 70 and 80

3. SALIENT CHARACTERISTICS

3.1 Dimensional and physical requirements Dimensional and physical requirements shall conform to Figure 1.

3.2 Material. Material shall be silicone rubber formulated and processed to meet the physical and mechanical properties specified in Table I of ZZ-R-765, when tested in accordance with the ASTM test method specified in Table IX of ZZ-R-765.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to Defense General Supply Center (DGSC), Attn. DGSC-VCA, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 9320

Distribution Statement A Approved for public release, distribution is unlimited

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3.3 Color Unless otherwise specified, the color of the silicone rubber shall be the natural color of the compound furnished.

3 4 Workmanship The end product shall be clean, smooth finished, and free from dirt, flash or rough edges.

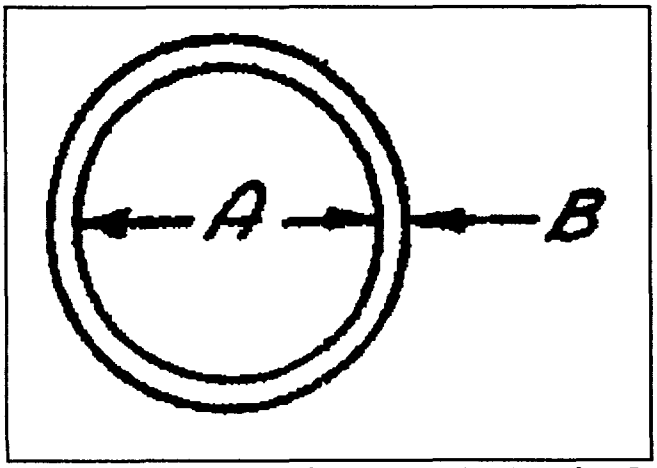



**Figure 1 Dimensional and Physical Requirements**



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Figure 1 continued

| PART NO. AAXXX09- | SPECIFIED SIZE | A INSIDE DIA. MAX./ MIN. | B WALL THICKNESS | TOLERANCES PLUS OR MINUS | TEMP. RATING OF | NONMETALLIC HARDNESS RATING | MIN. TENSILE STRENGTH | COLOR | LENGTH |
|---|---|---|---|---|---|---|---|---|---|
| 001 | 0 022 (No 24) | 0 027/0 020 | 0 012 | 0 002 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 010 | 0 027 (No 22) | 0 032/0 025 | 0 012 | 0 002 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 020 | 0 034 (No 20) | 0 039/0 032 | 0 016 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 030 | 0 042 (No 18) | 0 049/0 040 | 0 016 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 040 | 0 053 (No 16) | 0.061/0.051 | 0 016 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 050 | 0 066 (No 14) | 0 074/0.064 | 0 016 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 060F6 | 0 085 (No 12) | 0 091/0 081 | 0 016 | 0.003 | -94 to 400 | 55 0 to 65 0 | 1200 P S I | NOTE 4 | NOTE 5 |
| 070 | 0 095 (No 11) | 0.101/0.091 | 0 016 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOPTE 5 |
| 080F5 | 0 106 (No 10) | 0 112/0.102 | 0 016 | 0 003 | -94 to 400 | 45 0 to 55 0 | 1200 P S I | RED | NOTE 5 |
| 090D6 | 0 118 (No 9) | 0.124/0.114 | 0 020 | 0 003 | -80 to 425 | 55 0 to 65 0 | 650 P S I | RED | 28,000 |
| 100<br>110F5 | 0.133 (No. 8)<br>0 148 (No 7) | 0 141/0.129<br>0.158/0.144 | 0 020<br>0 020 | 0 003<br>0 003 | NOTE 2<br>-94 to 400 | NOTE 3<br>45 0 to 55 0 | NOTE 3<br>1200 P S I | NOTE 4<br>NOTE 4 | NOTE 5<br>NOTE 5 |
| 120 | 0 166 (No 6) | 0 178/0.162 | 0 020 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 130D5 | 0 186 (No 5) | 0 198/0 182 | 0 020 | 0 003 | -80 to 425 | 45 0 to 55 0 | 700 P S I | NOTE 4 | NOTE 5 |
| 131D5 | 0 186 (No 5) | 0.198/0.182 | 0 020 | 0 003 | -80 to 425 | 45 0 to 55 0 | 700 P S I | RED | NOTE 5 |
| 132D5 | 0 186 (No 5) | 0 198/0 182 | 0 020 | 0 003 | -80 to 425 | 45.0 to 55 0 | 700 P S I | GREEN | 15,000 |
| 133 C6 | 0 186 (No 5) | 0 198/0 182 | 0 020 | 0 003 | -80 to 425 | 55 0 to 65 0 | 650 P S I | GREEN | NOTE 5 |
| 140 | 0 208 (No 4) | 0 224/0 204 | 0 020 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 150 | 0 234 (No 3) | 0 249/0.229 | 0 020 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 160 | 0 263 (No 2) | 0.278/0.258 | 0 020 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 170 | 0 294 (No 1) | 0 311/0 289 | 0 020 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 180 | 0 330 (No 0) | 0 347/0 325 | 0 020 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 190 | 0 313 (5/16) | 0 334/0 312 | 0 025 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |





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Figure 1 continued

| PART NO. AAXXX09- | SPECIFIED SIZE | A INSIDE DIA. MAX. / MIN. | B WALL THICKNESS | TOLERANCES PLUS OR MINUS | TEMP. RATING OF | NONMETALLIC HARDNESS RATING | MIN. TENSILE STRENGTH | COLOR | LENGTH |
|---|---|---|---|---|---|---|---|---|---|
| 200C5 | 0 375 (3/8) | 0 399/0 375 | 0 025 | 0 003 | -80 to 425 | 45 0 to 55 0 | 700 P S I | NOTE 4 | NOTE 5 |
| 201C7 | 0 375 (3/8) | 0 399/0 375 | 0 025 | 0 003 | -80 to 425 | 65 0 to 75 0 | 650 P S I | NOTE 4 | NOTE 5 |
| 292D5 | 0 375 (3/8) | 0 399/0 375 | 0 025 | 0 003 | -80 to 425 | 45 0 to 55 0 | 700 P S I | NOTE 4 | NOTE 5 |
| 210 | 0 438 (7/16) | 0 462/0.438 | 0 025 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 220 | 0 500 (1/2) | 0 524/0 500 | 0 025 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 230 | 0 625 (5/8) | 0 655/0 625 | 0 030 | 0 003 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 240 | 0 750 (3/4) | 0 786/0 750 | 0 035 | 0 005 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 250 | 0 875 (7/8) | 0 911/0 875 | 0 035 | 0 005 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 260 | 1 000 (1) | 1 036/1 000 | 0 035 | 0 005 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 270 | 1 250 (1 1/4) | 1 290/1 250 | 0 040 | 0 005 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 280 | 1 500 (1 1/2) | 1 550/1.500 | 0 045 | 0 006 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 290 | 1 750 (1 3/4) | 1 812/1.750 | 0 055 | 0 008 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |
| 300 | 2 000 (2) | 2 070/2 000 | 0 060 | 0 010 | NOTE 2 | NOTE 3 | NOTE 3 | NOTE 4 | NOTE 5 |

NOTES:
1. Dimensions are in inches
2. Temperature rating is determined by the class of material specified per ZZ-R-765/GEN
3. Nonmetallic Hardness Rating (Shore Durometer A) and Minimum Tensile Strength are determined by the grade of material specified per ZZ-R-765/GEN.
4. Unless otherwise specified, the color will be the natural color of the material furnished
5. Shall be furnished in 100, 200, 500 or 1000 foot coils unless otherwise specified in the contract or purchase order Each shall contain not more than three individual lengths of tubing per 100 feet, and no individual length of tubing shall be less than 5 feet.



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4 REGULATORY REQUIREMENTS

4.1 Recovered materials The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR)

4 2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest version of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product

5. QUALITY ASSURANCE PROVISIONS

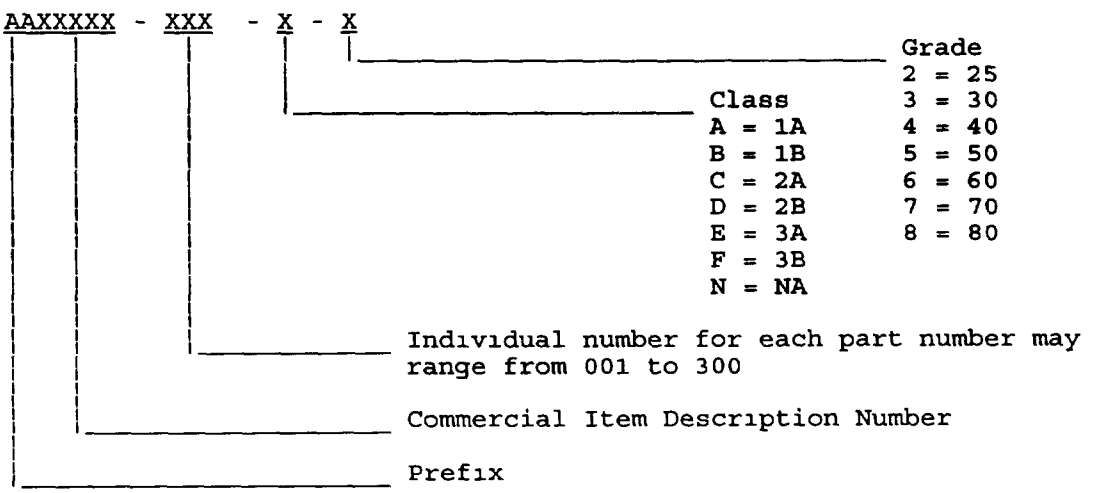
5.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided under the provisions of the contract.

5.2 Market acceptability. The item offered must have been sold to the Government or commercial market and reflect demonstrated reliability and serviceability

6 PRESERVATION, PACKAGING, PACKING AND MARKING. Preservation, packaging, packing and marking for shipment shall be in accordance with American Society for Testing and Materials (ASTM) D 3951 or as specified in the contract or order.

7 NOTES

7 1 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PINs to be used for items acquired to this description are created as follows:






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7 2 Addresses for obtaining reference documents

SAE

SAE
400 Commonwealth Drive
Warrendale PA 15096-0001

ASTM

American Society for Testing and Materials
1916 Race Street
Philadelphia PA 19103

7 3 Ordering data

a) Title, number, and date of this description
b) Class and grade required
c) Length.
d) Color required if other than natural color of compound furnished.
e) Specify any special packaging requirements.
f) Specify any additional quality assurance provisions, deemed necessary, covered by ZZ-R-765.
g) Contracting officer may request proof of certification of commercial item prior to first contract delivery.

7.4 Utilization caution. Use of this material in items that may come in contact with human skin, food, or potable water is not recommended without prior approval of the appropriate medical service

MILITARY INTEREST

CUSTODIANS
Army - MR
Navy - AS
Air Force - 11

CIVIL AGENCY COORDINATING ACTIVITY.

GSA - FSS

PREPARING ACTIVITY.

DLA - GS

(Project No : 9320-1116)



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